Algebra Essentials

ANSWER KEY

Section 1: Linear Equations & Distribution

1 If $3(2x - 4) = 4x + 6$, what is the value of x?

a. 9 ✓

b. 6

c. 8

d. 10

Expand the left side: $6x - 12 = 4x + 6$. Subtract $4x$ from both sides: $2x - 12 = 6$. Add 12: $2x = 18 \Rightarrow x = 9$.

2 If $\frac{x}{3} + \frac{x}{4} = 14$, what is the value of x?

a. 24 ✓

b. 21

c. 28

d. 18

Multiply the entire equation by 12: $4x + 3x = 168 \Rightarrow 7x = 168 \Rightarrow x = 24$.

3 If $2x + y = 17$ and $x - y = 4$, what is the value of xy?

a. 21 ✓

b. 18

c. 24

d. 27

Add both equations: $3x = 21 \Rightarrow x = 7$. Substitute into second equation: $7 - y = 4 \Rightarrow y = 3$. Thus, $xy = 7 \times 3 = 21$.

Section 2: Difference of Squares & Algebraic Identities

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4 If $x^2 - y^2 = 48$ and $x - y = 6$, what is the value of $x + y$?

a. 8 ✓

b. 6

c. 12

d. 14

Use the difference of squares identity: $x^2 - y^2 = (x - y)(x + y)$. Thus, $48 = 6(x + y) \Rightarrow x + y = \frac{48}{6} = 8$.

5 If $x + y = 10$ and $xy = 21$, what is the value of $x^2 + y^2$?

a. 58 ✓

b. 62

c. 54

d. 79

Square the sum: $(x + y)^2 = x^2 + 2xy + y^2$. Then $100 = x^2 + y^2 + 2(21) = x^2 + y^2 + 42$. Thus, $x^2 + y^2 = 100 - 42 = 58$.

6 Evaluate the numerical expression: $101^2 - 99^2$.

a. 400 ✓

b. 200

c. 800

d. 300

Apply difference of squares: $(101 - 99)(101 + 99) = 2 \times 200 = 400$.

Section 3: Inequalities & Absolute Values

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7 What is the solution set of $|2x - 5| = 9$?

a. $\{7, -2\}$ ✓

b. $\{7, 2\}$

c. $\{-7, 2\}$

d. $\{5, -4\}$

Case 1: $2x - 5 = 9 \Rightarrow 2x = 14 \Rightarrow x = 7$. Case 2: $2x - 5 = -9 \Rightarrow 2x = -4 \Rightarrow x = -2$. Solutions: $\{7, -2\}$.

8 If $-3x + 7 \leq -5$, what is the range of x?

a. $x \geq 4$ ✓

b. $x \leq 4$

c. $x \geq -4$

d. $x \leq -4$

Subtract 7: $-3x \leq -12$. Divide by -3 and reverse the inequality sign: $x \geq 4$.

9 If $x < y < 0$, which of the following expressions must be positive?

a. x^2y^2 ✓

b. $x - y$

c. $x + y$

d. $\frac{x}{y^3}$

Since both x and y are negative, $x^2 > 0$ and $y^2 > 0$, so their product x^2y^2 is strictly positive.

Section 4: Advanced Systems & Substitution Tricks

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10 If $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$ and $xy = 12$, what is the value of $x + y$?

a. 10 ✓

b. 12

c. 8

d. 15

Combine fractions: $\frac{x+y}{xy} = \frac{5}{6}$. Since $xy = 12$, we have $\frac{x+y}{12} = \frac{5}{6} \Rightarrow x + y = 12 \times \frac{5}{6} = 10$.

11 If $a + b + c = 18$ and $a + b = 11$, while $b + c = 13$, what is the value of b?

a. 6 ✓

b. 5

c. 7

d. 8

From the first and second equations: $c = 18 - 11 = 7$. From $b + c = 13$, we get $b = 13 - 7 = 6$.

12 If $x^2 + 6x + 9 = 25$ and $x > 0$, what is the value of x?

a. 2 ✓

b. 3

c. 4

d. 5

Factor as a perfect square: $(x + 3)^2 = 25$. Since $x > 0$, $x + 3 = 5 \Rightarrow x = 2$.

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